C4.5 What happens to products at the end of their useful life?	
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*
Iron is the most widely used metal in the world. The useful life of products made from iron is limited because iron corrodes. This involves an oxidation reaction with oxygen from the air. Barrier methods to prevent corrosion extend the useful life of metal products, which is good for consumers and has a positive outcome in terms of the life cycle assessment. Sacrificial protection uses a more reactive metal such as zinc to oxidise in preference to iron. This continues to prevent corrosion even if the coating on the metal is damaged. Life cycle assessments (LCAs) are used to consider the overall impact of our making, using and disposing of a product. LCAs involve considering the use of resources and the impact on the environment of all stages of making materials for a product from raw materials, making the finished product, the use of the product, transport and the method used for its disposal at the end of its useful life. It is difficult to make secure judgments when writing LCAs because there is not always enough data and people do not always follow recommended disposal advice (IaS4). Some products can be recycled at the end of their useful life. In recycling, the products are broken down into the materials used to make them; these materials are then used to make something else. Reusing products uses less energy than recycling them. Reusing and recycling both affects the LCA.	1. describe the conditions which cause corrosion and the process of corrosion, and explain how mitigation is achieved by creating a physical barrier to oxygen and water and by sacrificial protection *(separate science only)* 2. explain reduction and oxidation in terms of loss or gain of oxygen, identifying which species are oxidised and which are reduced **3. explain reduction and oxidation in terms of gain or loss of electrons, identifying which species are oxidised and which are reduced** 4. describe the basic principles in carrying out a life-cycle assessment of a material or product including a) the use of water, energy and the environmental impact of each stage in a life cycle, including its manufacture, transport and disposal b) incineration, landfill and electricity generation schemes c) biodegradable and non-biodegradable materials 5. interpret data from a life-cycle assessment of a material or product

Linked learning opportunities
Practical work: • Investigating the factors needed for rusting of iron or corrosion of other metals. • Investigating the effectiveness of corrosion prevention (barrier and sacrificial protection methods). **Ideas about Science:** • use the example of applying scientific solutions to the problem of corrosion of metals to explain the idea of improving sustainability (IaS4) • use life cycle assessments to compare the sustainability of products and processes (IaS4)

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Recycling conserves resources such as crude oil and metal ores, but will not be sufficient to meet future demand for these resources unless habits change. The viability of a recycling process depends on a number of factors: the finite nature of some deposits of raw materials (such as metal ores and crude oil), availability of the material to be recycled, economic and practical considerations of collection and sorting, removal of impurities, energy use in transport and processing, scale of demand for new product, environmental impact of the process. Products made from recycled materials do not always have a lower environmental impact than those made from new resources (IaS4).	6. describe the process where PET drinks bottles are reused and recycled for different uses, and explain why this is viable 7. evaluate factors that affect decisions on recycling with reference to products made from crude oil and metal ores	